
Emergent Communication through Deep Learning

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Abstract

Emergent communication explores how communication protocols arise among interacting agents in multi-agent systems through task learning. In this paper, we assess a recent method, Metropolis-Hastings Naming Game (MHNG), in emergent communication using the EGG toolkit for simulating emergent communication. We replicate and evaluate key findings from the MH Naming game study and compare EGG’s approach with the other methodology in the literature. We identify trade-offs between task efficiency, generalizability, and interpretability in emergent protocols. Experiment results show that EGG achieves lower reconstruction loss, while MHNG produces more interpretable communication patterns. Future directions include extending emergent communication to more complex datasets, multimodal inputs, and larger agent populations to advance AI communication systems.

1 Introduction

Emergent communication investigates how communication systems arise among interacting agents, driven by cooperative tasks and optimization needs. Frameworks like **EGG** (Kharitonov et al., 2019) have been pivotal in advancing this research. EGG provides a modular toolkit for simulating communication games, optimizing channels using Gumbel-Softmax relaxation or REINFORCE, and analyzing emergent protocols. By supporting both single-symbol and variable-length messages, EGG enables researchers to explore how communication evolves in structured environments.

However, significant challenges remain. Emergent communication protocols often deviate from human linguistic behavior, as agents lack cognitive constraints such as memory limitations, role alternation, and compositionality (Chaabouni et al., 2022). Addressing these limitations within EGG could help produce protocols that better align with natural languages. Another issue is generalization: emergent systems struggle to transfer learned communication to novel tasks, such as numerical reasoning (AAAI, 2024). EGG’s flexibility makes it well-suited for addressing these gaps through benchmark tasks and extended evaluation scenarios. Moreover, scalability also poses challenges for emergent communication. Research shows that as group sizes grow, communication consistency decreases, even though task performance remains stable (NeurIPS, 2021). EGG provides a robust environment for studying how communication protocols evolve in large, complex agent populations and varying network structures.

Emergent communication has broader implications for multi-agent reinforcement learning (MARL) and natural language processing (NLP). By facilitating task-specific communication strategies, MARL systems achieve improved coordination in domains like robotics and game simulations. Recent work suggests combining emergent communication with supervised NLP can enhance AI naturalness and interpretability (arXiv, 2023). EGG’s modular design positions it as an ideal tool to integrate these insights and explore new research directions.

Future work with EGG should focus on incorporating cognitive constraints, designing benchmarks for compositional generalization, and analyzing large-scale agent dynamics. By addressing these

challenges, EGG can further our understanding of language emergence and contribute to building scalable, interpretable, and human-like AI systems.

2 Method

2.1 Emergence of lanGuage in Games (EGG)

EGG framework (Kharitonov et al., 2019) is a modular and open-source toolkit for studying emergent communication in cooperative games. EGG supports flexible game implementation, offering optimization techniques like REINFORCE and Gumbel-Softmax to train agents’ communication channels. These techniques enable agents to optimize their signals for a given task, with REINFORCE addressing discrete signals via reinforcement learning and Gumbel-Softmax allowing differentiable approximations of discrete signals.

In our study, we replicate the MNIST Autoencoder Game, where two agents—a Sender and a Receiver—are trained to communicate for image reconstruction. This game tasks a Sender and a Receiver agent to develop a communication protocol to reconstruct images from the MNIST dataset. Specifically, the Sender observes an input MNIST image, encodes it into a message (either discrete or continuous), and transmits it to the Receiver. The Receiver then decodes the message to reconstruct the original image. The reconstruction quality is evaluated using a binary cross-entropy loss.

The loss function is defined as follows:

$$\mathcal{L}_i = -\frac{1}{N} \sum_{j=1}^N [x_{ij} \log(\hat{x}_{ij}) + (1 - x_{ij}) \log(1 - \hat{x}_{ij})]$$

where:

- x_{ij} is the j -th pixel of the original input image x_i ,
- $\hat{x}_{ij}$ is the j -th pixel of the reconstructed image $\hat{x}_i$,
- N is the total number of pixels in the image (here $N = 784$).

The mean loss over a batch of M samples is then given by:

$$\mathcal{L} = \frac{1}{M} \sum_{i=1}^M \mathcal{L}_i$$

This loss measures the reconstruction error between the original image x and the reconstructed output $\hat{x}$, where a lower value indicates better reconstruction. The game is trained over 50 epochs with a batch size of 10, and the loss is monitored to analyze the agents’ communication effectiveness.

2.2 Metropolis-Hastings Naming Game

The Metropolis-Hastings Naming Game is a probabilistic framework used to study symbol emergence among interacting agents. In this approach, agents collectively develop shared symbols or categories for observed stimuli without explicit supervision. In the study by Taniguchi, T. et al. (2023), each agent is modeled using a Variational Autoencoder (VAE) for latent representation learning and a Gaussian Mixture Model (GMM) to cluster the latent space. Agents interact through a naming game, where the goal is to assign consistent symbols to observed data.

The communication process is facilitated using the Metropolis-Hastings algorithm, which iteratively updates the agents’ symbol assignment probabilities based on agreement among agents. The loss function in this study focuses on minimizing the discrepancy between agents’ symbol assignments and the consistency of categories in the shared latent space. The MNIST dataset is used as input, where agents develop categories (symbols) for the handwritten digits.

For evaluation, the study employs the MNIST dataset with 10 categories and 10,000 samples, training two agents (Agent A and Agent B) over 50 epochs with a batch size of 10. Each agent’s performance

is assessed using the Adjusted Rand Index (ARI), which measures the agreement between the agent’s symbol clusters and the true MNIST labels. The Kappa coefficient evaluates the agreement of symbols (w^A and w^B) between the two agents. Additional metrics include the number of successful symbol exchanges: A2B (Agent A to Agent B) and B2A (Agent B to Agent A).

2.3 Comparison and Evaluation

To compare the **MNIST Autoencoder Game (EGG)** and the **MH Naming Game (MHNG)**, we evaluate their performance on the MNIST dataset:

1. **EGG**: Performance is measured using the binary cross-entropy reconstruction loss, averaged over the dataset. Lower loss indicates better communication between the Sender and Receiver for reconstructing images.
2. **MHNG**: Evaluation focuses on symbol consistency and category agreement using ARI and Kappa coefficients. High ARI values indicate that agents’ emergent symbols align with true MNIST categories, while high Kappa reflects strong inter-agent agreement.

By comparing EGG’s reconstruction loss with MHNG’s ARI and Kappa scores, we aim to highlight the differences in evaluation goals: EGG prioritizes task-specific performance (image reconstruction), while MHNG emphasizes the interpretability and consistency of emergent symbols.

3 Experiments

3.1 Dataset and Setup

In this experiment, we use the MNIST dataset to reproduce the MH naming game using the EGG toolkit.

3.1.1 MNIST Dataset

The MNIST dataset is a benchmark dataset consisting of 28×28 grayscale images of handwritten digits (0–9), with 60,000 training samples and 10,000 test samples. Each image is represented as a 784-dimensional vector, with pixel values normalized to [0, 1]. Due to its simplicity and standardized format, MNIST is widely used for validating machine learning models, including classification networks, autoencoders, and emergent communication frameworks like EGG.

In our study using the EGG framework, we implement a Vision module to encode MNIST images into a 500-dimensional feature vector. This module consists of convolutional and fully connected layers that extract meaningful latent representations of the images. To pre-train the Vision module, we define an auxiliary classification task, where a PretrainNet model classifies MNIST digits into 10 classes (0–9) using the Vision module’s output. By minimizing the classification loss on the MNIST training set, the Vision module learns robust image features. These pre-trained features are then used in the MNIST autoencoder game within the EGG framework, where agents communicate to reconstruct input images effectively. This pre-training ensures a strong initialization, improving downstream communication and reconstruction performance.

3.1.2 Game Setup

The communication between agents is facilitated through a discrete communication channel. The experiment compares two reinforcement learning-based methods with the Metropolis-Hastings method:

1. **MH Naming Game**: Communication is based on the Metropolis-Hastings algorithm. Each agent iteratively refines its internal representation by sampling signs and updating them based on a probabilistic acceptance ratio.
2. **Reinforce-Based One-Symbol Communication**: Communication occurs using a single discrete symbol. Agents are optimized using the Reinforce algorithm, which estimates gradients of the expectation of the loss using the log-probability trick.
3. **Reinforce-Based Variable-Length Communication**: Communication involves sequences of symbols generated by RNNs, terminated by an end-of-sequence (EOS) symbol.

3.2 Metropolis-Hastings Communication Design

The MH Naming Game uses a probabilistic communication mechanism. The process operates iteratively as follows:

- Each agent proposes a sign w sampled based on its current internal representation z .
- The receiving agent calculates an **acceptance ratio** r based on the proposal and its own internal representation. The acceptance ratio is defined as:

$$r = \min \left(1, \frac{P(z^B|w^A)}{P(z^A|w^B)} \right), \quad (1)$$

where P represents the probability of the observed data given the sign.

- The receiving agent accepts or rejects the sign probabilistically according to r .

3.3 EGG Toolkit Design

The EGG toolkit simplifies the implementation of communication games through modular design. EGG operates over four main components:

- **Sender and Receiver:** The two communicating agents. Sender generates messages, and Receiver interprets them.
- **Game:** Orchestrates the flow of data between agents, computes the loss, and facilitates training.
- **Trainer:** Optimizes the agents' parameters, manages checkpoints, and supports early stopping.

EGG supports both single-symbol and variable-length communication through predefined wrappers:

- Single-symbol: `core.ReinforceWrapper`.
- Variable-length: `RnnSenderReinforce` and `RnnReceiverReinforce`.

3.4 Network Architecture and Training

The Sender and Receiver networks are defined as follows:

- **Sender:** A convolutional neural network (CNN) processes the MNIST image into a latent representation. For one-symbol communication, the latent vector is passed through a softmax layer to produce a probability distribution over the vocabulary. For variable-length communication, the latent representation initializes the hidden state of an RNN that generates a message sequence.
- **Receiver:** For one-symbol communication, the Receiver processes the message embedding to classify the digit. For variable-length communication, the Receiver processes the symbol sequence using an RNN and generates the final output.

After training, we visualize the results by comparing the original input image and the Receiver's reconstructed output.

3.4.1 Hyperparameters

The agents are trained using Reinforce for 50 epochs to ensure consistent comparison with the MH Naming Game. It is trained with the following hyperparameters:

- **Optimizer:** Adam with learning rate 0.001.
- **Vocabulary Size:** 10 symbols (including the EOS marker).
- **Hidden Size (RNN):** 64.
- **Max Sequence Length:** 5 symbols.
- **Batch Size:** 32.

3.4.2 Hardware and Environment

The training is conducted on a CUDA-compatible GPU if available; otherwise, it defaults to the CPU. The entire training pipeline is initialized and optimized using PyTorch and EGG-level arguments.

4 Results



Figure 1: Reinforce-based communication outputs: the top row shows original inputs, and the bottom row shows reconstructions.

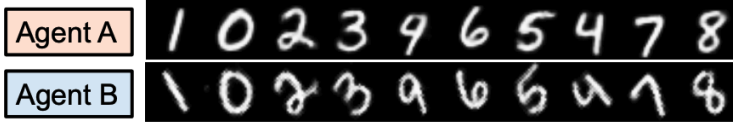


Figure 2: MH Naming Game: Agent A observes raw digits, while Agent B observes corresponding rotated digits.

Table 1: Final Epoch Loss Results for EGG Games

Method	Final Epoch Loss
Reinforce (One Symbol)	1.1503
Reinforce (Variable Length)	1.2380

5 Comparison

As shown above in Figure 1 and Figure 2, both the Reinforce-based algorithm and the Metropolis-Hasting algorithm have a successful communication process, where the reconstructions (bottom row) closely resemble the original digits (top row). The results presented in Table 1 indicate that the one-symbol communication achieves a lower final epoch loss of 1.1503 compared to the variable-length communication, which incurs a loss of 1.2380. The lower loss for single-symbol communication suggests that it is more efficient in conveying concise information with reduced complexity, while variable-length messages offer greater flexibility for richer symbol expressions, but with slightly higher computational cost.

By comparing the final output of both algorithms, MH algorithm has a more clear reconstruction output of the original digits than Reinforce-based algorithm but has a relatively higher epoch loss. This highlights the difference between two approaches: EGG’s Reinforce-based communication prioritizes performance accuracy by optimizing its loss and is effective in performance-driven or task-based communication, while the MH algorithm emphasizes more unsupervised category formation without explicit rewards and emergence of biologically plausible symbols.

6 Conclusion & Discussion

The study of emergent communication in deep learning agents has gained significant momentum in recent years, driven by the dual goals of understanding human language evolution and developing robust, interpretable communication systems for artificial agents. In this project, the critical assessment of different frameworks in emergent communication is structured, focusing on their effectiveness, limitations, and alignment with theoretical insights from cognitive science and linguistics.

This project explores the usage of EGG framework in emergent communication by conducting a result replication of Metropolis-Hastings framework and using the MNIST dataset as the benchmark. The EGG framework offers a robust platform for simulating communication between neural agents in Sender-Receiver games. It has particular advantages in supporting different optimization strategies like Reinforce-based and Gumbel-Softmax relaxation, while at the same time, it is capable for efficient task-oriented communication with loss optimization.

While EGG framework has successfully made a similar output with Metropolis-Hastings framework through a Reinforce-based communication in this project, it also shows some limitations when comparing to Metropolis-Hastings framework, such as less interpretable results. The key distinction between the two frameworks lies in their objectives and methodologies. EGG’s Reinforce-based communication has specific redefined task as a primary goal, the MH algorithm offers a more cognitively inspired approach with no explicit rewards or supervision. This highlights the different applications in practical scenarios of these two frameworks: EGG is ideal for rapid studying task-oriented communication or in the cases where communication serves an immediate functional purpose, while MH provides a theoretical framework for understanding human-like language evolution and symbol formulation in naturalistic and ambiguous environments.

Although the EGG framework remains a powerful platform for exploring emergent communication, there is a need to move beyond task-specific optimization toward frameworks that integrate efficiency, generalizability, and interpretability. Looking forward, the results of this project point to some possible directions in future researches. For example, integrating the optimization strengths of EGG framework with the probabilistic inference principles of MH framework could yield systems that are both task-efficient and capable of learning robust & human-like communication. More specifically, combining cognitive constraints, such as joint attention and category alignment from the MH framework, with the EGG framework could improve the interpretability and robustness of emergent protocols. Additionally, extending the frameworks to larger datasets and more complex tasks, such as numerical reasoning or multi-agent coordination, could provide further insights into emergent communication. Real-world tasks with multimodal inputs and more sophisticated environments will further evaluate their scalability and generalization capabilities.

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